DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
UNIVERSITY OF BARISHAL
FINAL EXAMINATION
Course Title: Mathematical Analysis for Computer Science
Course Code: CSE-3201
3rd Year 2nd Semester
Session: 2019-20(Admission)
(Answer Any Five Questions)

Time: 3 Hours

Marks: 60

1. a) Write down the reason of studying mathematical analysis in Computer Science and Engineering. [3]

   b) Derive the recursive formula for the minimum number of moves required to solve the Tower of Hanoi problem with n disks. [3]

   c) How many intersection points are there when n distinct lines are drawn in the plane, assuming no two lines are parallel and no three lines meet at the same point? [3]

   d) Derive the recursive formula for the position of the survivor in the Josephus problem, given that every k-th person is eliminated from a circle of n people. [3]

2. a) Define Box and Whisker Plot. Set Q contains 20 positive integer values. The smallest value in Set Q is a single digit value and the largest value in Set Q is a three digit value. What are possible values for the range of Set Q? Find out the minimum and maximum range for set Q.[Range in a set means, $Range_Q = Q_{max} - Q_{min}$.] [3]

   b) Solve the recurrence relation T(n)=T(n−1)+n with T(1)=1. What is the closed-form solution for T(n)? **Hint:** This is the sum of the first n natural numbers. Use the formula for the sum of an arithmetic series: $S(n) = \frac{n(n+1)}{2}$ [3]

   c) To prove the statement: $m = \lfloor log_2(n) \rfloor + 1$ where n is an m-bit integer, we need to relate the number of bits required to represent n in binary to the logarithm of n. [4]

   d) If 724 is the greatest common divisor of positive integers A and B, and 726 is the least common multiple of A and B, then AB=? [2]

3. a) A number in decimal notation is divisible by 3 if and only if the sum of its digits is divisible by 3. Prove this well-known rule, and generalize it. [2]

   b) Define relative primality. The formula for the n-th **Catalan number** is: $C_n = \frac{1}{n+1}\binom{2n}{n}$ calculate C_3? [4]

   c) Define bayes theorem. What do you mean by priori and posterior probability. A bag I contains 4 white and 6 black balls while another Bag II contains 4 white and 3 black balls. One ball is drawn at random from one of the bags, and it is found to be black. Find the probability that it was drawn from Bag I. [3]

   d) What do you mean by complex numbers? Prove that √2 is irrational number. [3]

4. a) Draw the transition diagram for each case. Is it irreducible? For both yes or no, each cases justify your answer with proper explanation. [4]

$$\text{(a) } P = \begin{pmatrix} 0.5 & 0.5 \\ 0.5 & 0.5 \end{pmatrix} \quad \text{(b) } P = \begin{pmatrix} 0.5 & 0.5 \\ 1 & 0 \end{pmatrix} \quad \text{(c) } P = \begin{pmatrix} 1/3 & 0 & 2/3 \\ 0 & 1 & 0 \\ 0 & 1/5 & 4/5 \end{pmatrix}$$

$$\text{(d) } P = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad \text{(e) } P = \begin{pmatrix} 1/2 & 1/2 & 0 \\ 0 & 1/2 & 1/2 \\ 1/3 & 1/3 & 1/3 \end{pmatrix}$$

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